

DETECTION OF FAKE NEWS USING HYBRID NLP AND ML

Declaration

I can state that the following dissertation is the work of my independent study and research, and its name is Detection of Fake News Using Hybrid NLP and Machine Learning Approaches. Any information, data and literature utilized has been properly recognized and quoted in line with academic conventions. This is not a previously submitted, in whole or part, academic award or qualification.

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Abstract

The uncontrolled development of misinformation and fake news on the Internet is a serious social and technological issue. This study explores the hybrid model that combines Natural Language Processing (NLP) and Machine Learning (ML) methods to efficiently detect fake news, and in particular, sarcastic and misleading headlines. The project uses secondary datasets that consist of more than 60,000 labeled news headlines to apply a systematic pipeline with data preprocessing, feature extraction with TF-IDF and classification with Logistic Regression, Support Vector Machine (SVM) and Random Forest. Model performance is measured on the basis of standard performance measures such as accuracy, precision, recall, F1-score and ROC-AUC. The SVM classifier was the most suitable model that was tested because it exhibited the optimal compromise between generalization and accuracy. Classification reliability was also determined using confusion matrices and ROC curves. The study also brings out the benefits of using statistical text representations and strong machine learning classifiers to identify subtle textual patterns that can be used to point out fake or sarcastic digital content. The paper ends with a conclusion about the scalability of the system to multilingual adaptation and real-time implementation. Comprehensively, the project can be seen as part of the expanding body of research on automated misinformation detection due to its illustration of a practical, interpretable, and expandable hybrid model of fake news detection in a digital ecosystem.

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Chapter 1. Introduction

Spreading the misinformation, often called in new-fangled jargon the fake news, has become a burning socio-conservative issue in the digital age. Printing and broadcasting have been gradually replaced by social media, where it has become difficult to control the flow of information that contributes more to the uncritical thinking of people and the way we vote and trust our society. In the wake of systems like traditional verification methods showing to be inadequate when it comes to scale, automated forms of detecting deceptive content have received more and more academic and practical attention.

Two aspects that can be used to solve this problem are Natural Language Processing (NLP) and Machine Learning (ML), which are two closely connected fields. NLP makes it possible to computationally analyse human speech when the syntactic, semantic, and contextual characteristics are taken out of textual information. Instead, ML enables the systems to be trained on such characteristics and make predictions. Related to the application of fake news detection, this has been defined as training models to discriminate between factual and fabricated content based on patterns extrapolated using labelled corpora.

More recent developments of hybrid systems, such as the combination of both traditional machine learning types (e.g., Support Vector Machines, Random Forests) and deep learning (e.g., Long Short-Term Memory networks, Bidirectional Encoder Representations from Transformers [BERT]), have shown improved performance. The use of NLP allows these models to exploit the richness in the language and generalisation abilities of machine learning classifiers.



In this dissertation, we explore the design, implementation and testing of a hybrid NLP-ML model that detects fake news. It includes preprocessing of data, engineering features with Term Frequency-Inverse Document Frequency (TF-IDF) and the contextual embeddings, choice of model, and possible performance evaluation comparison (Nadeem *et al.*, 2024). It also considers weaknesses of the prior art practices and proposes an architecture that is better both in terms of linguistic detail and classification strength. Dissertation organisation has a systematic structure of theoretical slant, including the methodology of its implementation, experimental verification and reflection on it.

Chapter 2: Aim, Objectives, and Rationale

2.1 Research Aim and Objectives

The ultimate goal to be achieved in this study is a hybrid model based on a combination of Natural Language Processing (NLP) techniques with Machine Learning (ML) algorithms to find a viable tool in detecting fake news in online, textual information. With the spread of false information through online services and social networks intensifying, there is an ever greater need to develop efficient automated solutions that will be able to detect the difference between genuine and fake news. NLP and ML synergy. Hybridisation of NLP and ML provides a synergistic model that can use both linguistic and statistical elements to find patterns in texts with datasets; therefore, it can better classify and stabilise the model.

To attain the mentioned goal, the following objectives have been developed:

1. To do an overview of the current fake news detection methods, with the emphasis on the conventional ML classifiers, deep learning architectures, and the latest NLP embeddings.
2. To clean and vectorise a benchmark fake news collection with the help of NLP algorithms, e.g., TF-IDF, Word2Vec, and contextual representation embeddings, like BERT.
3. To build and train various models of machine learning, such as Logistic regression, Support Vector Machines(SVM) models, Random Forests and Long Short-term Memory (LSTM) models based on extracted features.

4. To adopt a model of hybrid detection that seeks to treat NLP-based feature extraction and conventional and deep learning-based classifiers to test the difference in performance.
5. To contrast the performance of the hybrid model based on the evaluation metrics of accuracy, precision, recall, F1-score and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) and compare it with baseline models.

In an attempt to enhance the practice of automated misinformation detection, this work combines both the semantic sensitivity of the NLP human language technology with the predictive capabilities of ML to address the drawback of the unimodal methods.

2.2 Research Rationale

Fake news has become a major socio-technical challenge around the use of digital ecosystems and social media platforms, in particular. It has been illustrated that misinformation campaigns can affect political performances, interfere with health campaigns and cause civil unrest. With the need to eliminate the inconvenience and even the harmfulness of such a complicated problem, researchers and technologists have tried to use artificial intelligence (AI), primarily Natural Language Processing (NLP) and Machine Learning (ML), to automate software modules that identify and filter misleading content. Yet, the current solutions tend to have limitations in flexibility, generalisation, and the understanding of the context.

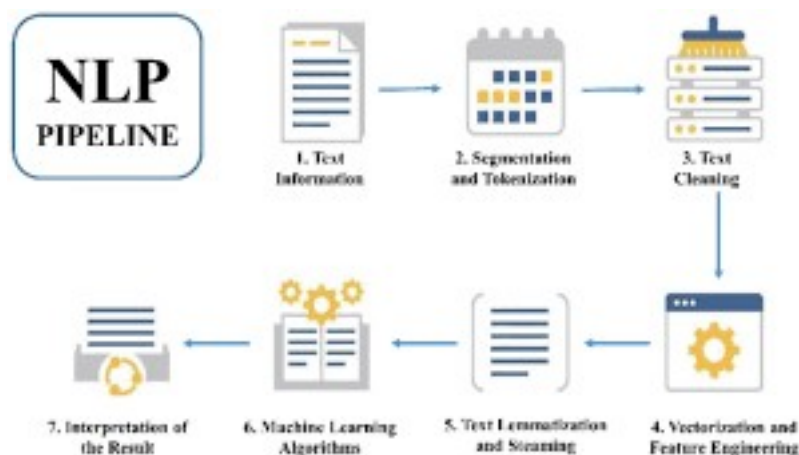


Figure 2 Fake news detection using NLP pipeline

The most common traditional machine learning-based techniques used in text classification tasks are Naive Bayes, Decision Trees, and Support Vector Machines (SVM). Such techniques rely too strongly on expert-crafted features that are unlikely to model the semantic complexity

of natural language effectively. Further, the strong dependency on Bag-of-Words (BoW) or Term Frequency-Inverse Document Frequency (TF-IDF) representations also brings limits to such poor representations since they are sparse and context-free. e.g. TF-IDF can be used to calculate the significance of term t in document d of corpus D with:

$$TF-IDF(t, d, D) = tf(t, d) \times \log\left(\frac{N}{|\{d \in D : t \in d\}|}\right).$$

Although the TF-IDF performs well to capture word significance, it does not consider syntax and semantics, and hence, when applied to a complex text, poor generalisation occurs.

Recurrent Neural Networks (RNNs), especially Long Short-Term Memory (LSTM) networks, proved to work better due to the preservation of word sequence. However, these models are computationally expensive, and they frequently experience vanishing gradients and poor memory when working with long textual data (Khanam *et al.*, 2021, March). Transformer-based models, including BERT (Bidirectional Encoder Representations from Transformers), overcome these drawbacks by characterising self-attention in the pursuit of understanding bidirectional context, improving syntactic/semantic relationship understanding of the model.

However, standalone models (ML-based or NLP-based) have not always performed well, especially in terms of accuracy, when tested in the real world, because of such problems as domain adaptation, feature sparsity, and class imbalance. It can be assumed that a hybrid uniform method involving feature engineering based on NLP and various ML classification methods could provide better generalisation (typical overtures of fake news, such as sensational language, subjective tones, and logical fallacies, could be detected).

In this research paper, the author posits the concept of the layered hybrid framework, where:

- Cleaning and organisation of textual inputs are carried out using NLP techniques, i.e. lemmatisation, named entity recognition, and dependency parsing.
- Such feature extraction is conducted on both traditional (TF-IDF) and modern (BERT) types of vectorisations.
- SVM, Random Forest and LSTM classification models are trained with stratified cross-validation to avoid over-fitting.
- Evaluation measures, e.g. precision $P = \frac{TP}{TP + FP}$, recall, and F1-score $F1 = 2 \times \frac{P \cdot R}{P + R}$, will be computed to assess model performance.

The hybrid architecture also seeks to fill the divide between the handcrafted statistical models and the data-hungry deep-learning systems by making use of their complementary advantages. This setting of integration is further explained based on recent scholarly publications, which point to the fact that a hybrid approach is consistently more accurate than a single approach in classification problems that involve unstructured or noisy text.

This dissertation will bring value to both fields: theoretical and applied, by participating with an empirical assessment of hybrid NLP-ML systems and studying the architecture of designing scalability. It fills in the gaps that still exist in terms of methodology and lays out a scheme that can be duplicated for similar tasks like hate speech detection, sentiment classification, or rumour analysis.

Chapter 3: Literature Review and Justification

3.1 Introduction to Fake News Detection

Fake news is the publication of false or misleading information in the name of a regular news publication. This effect is a great risk to the integrity of digital information ecosystems, especially when applied to the case of social media platforms where content spread is high and mostly unmitigated. The spread of this kind of misinformation is therefore an urgent source of concern both within computational and social scopes because it can not only affect the mass opinion, but also play a manipulative role within the political sphere, and cause social turmoil.

Technically, detecting fake news is quite a complex task. Constructed articles usually resemble the architecture and semantics of actual news, which makes the automatic distinction more complicated. There is also the dynamism of the language in use and ever-changing tactics of misinformation agents, which require dynamic and situation-specific solutions (Prachi *et al.*, 2022). More detailed syntactic and semantic variations cannot be captured only with traditional keyword-based filters or rule-based systems, and more advanced detection mechanisms require more attention.

[illegible]

Considering the extent of content creation on the internet, it is necessary to adopt robust and scalable hybrid models based on NLP and ML, which scale well with a large amount of text and carry out a classification and understanding of text. This chapter describes the literature that is relevant with regard to the domain of interest, which offers the base of the proposed hybrid model that can fulfil the current limitations and enhance the accuracy of detection.

3.2 Natural Language Processing (NLP) Techniques

The Natural Language Processing (NLP) is one of the most critical areas in the manipulation of human language, and its understanding computationally. When applied to spotting fake news, NLP enables the processing of unstructured textual information into a structured numerical representation, which can be engaged with the help of machine learning and deep learning models. In this section, the authors describe traditional and modern NLP methods concerning textual feature representation, their architecture, their shortcomings, and their suitability to hybrid modelling in fake news systems.

3.2.1 Bag of Words (BoW)

The Bag of Words model is one of the techniques that are at the core of NLP since it reduces the text to a list of words, ignoring the grammar, word order, and meaning. The main idea here is to build the vocabulary of known words out of a corpus and to view each document as a frequency-based vector. Mathematically, in case the vocabulary is $V = \{w_1, w_2, \dots, w_n\}$, to represent the data, the documents d are converted into a vector $v_d \in R_n$, with each element $v_{d,i}$ provided as the frequency of word w_i in the document d . However, even in its simplest form, BoW tends to produce high-dimensional, sparse feature spaces and overfits in situations where semantic generalisation is of concern, as in the case of complex fake news detection.

3.2.2 Term Frequency–Inverse Document Frequency (TF-IDF)

TF-IDF is an improvement of each word resulting in BoW as it mixes the level of importance of terms at a corpus level with a balance of frequency by term in a document and by inverse frequency across the whole corpus. The Term Frequency (TF) is explained to be:

$$TF(t, d) = \frac{f_{t,d}}{\sum_k f_{k,d}}$$

With $f_{t,d}$ the frequency of term t in document d , and the denominator is a sum over all term frequencies in that document.

To make up for this commonality of the common words, there is the Inverse Document Frequency (IDF):

$$IDF(t, D) = \log\left(\frac{N}{1 + |\{d \in D : t \in d\}|}\right)$$

Here, N is the total number of documents, and the denominator represents a count of documents that contain the term t. The last TF-IDF is provided as:

$$TF - IDF(t, d, D) = TF(t, d) \cdot IDF(t, D)$$

TF-IDF is a comfortable, interpretable feature in shallow models, but it is not able to capture syntactic dependencies or semantic relationships, which tend to play a crucial role in the identification of subtle misinformation.

3.2.3 Word Embeddings: Word2Vec and GloVe

Word embeddings, in contrast to sparse representations of vectors, represent words in dense, continuous vectors, in which semantically similar words are placed in close positions. The principle according to which they are trained is called distributional semantics, i.e., the hypothesis that words with similar contexts can be assumed to be related to the same meaning.

Word2Vec, presented with 2 basic architectures, namely Continuous Bag of Words (CBOW) and Skip-Gram, relies on a shallow neural network to capture its vector representations. An example is that of the Skip-Gram model that maximises the probability of context words $w_{context}$

given a target word w_t as:
$$P(w_{context} | w_t) = \frac{\exp(v_{w_t} \cdot v_{w_{context}})}{\sum_{w \in V} \exp(v_{w_t} \cdot v_w)}$$

In this case, v is the word w embedding vector, and V is the set of vocabulary.

In comparison, GloVe (Global Vectors) is also a count-based model, which is built around a co-occurrence matrix, which is then factorised to find word vectors. The criterion that is minimised in the model is a weighted least squares objective:

$$J = \sum_{i,j=1}^{|V|} f(X_{ij}) (v_i^T \sim v_j + b_i + b_{\sim j} - \log X_{ij})^2$$

Where the X_{ij} is the count of the occurrence of word j in the context of word i , the vectors used are v_i and $\{\sim v_i\}$ which are the word and context word vectors, respectively, and f is the kind of weighting that is being done.

Such embeddings represent quite complex and meaningful patterns of semantics like analogical relations (e.g., king - man + woman - queen), and they are appropriate when being chosen as input for deep learning models.

3.2.4 Transformers and Contextual Embeddings

Whereas traditional embeddings would have a one-time set/static vector per word, with the advent of transformer-based models, the concept of contextual embedding made its way: since surrounding words affect the meaning of words, the embeddings would vary depending on the words next to it (Alghamdi *et al.*, 2022). It is crucial when the interpretation of words is context-sensitive, since it is necessary to disambiguate a polysemous word in fake news.

BERT (Bidirectional Encoder Representations from Transformers) employs a bidirectional attention scheme that allows it to infer what a word means by glancing both left and right. It is built upon the Transformer encoder architecture, where it calculates:

$$Attention(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

In which Q, K, V are query, key and value matrices, and d_k is the dimension against which key vectors are dimensioned.

RoBERTa is a variation of BERT that has even further tightened the performance by the removal of the next sentence prediction task as well as training over larger datasets using dynamic masking.

In a nutshell, NLP methods have been developed from a frequency model such as the BoW and TF-IDF to sophisticated semantic-based architectures such as Word2Vec, GloVe, and BERT. Every approach has its advantages and disadvantages. The hybrid design of contextual embeddings integrated with robust classifiers is the core of hybrid models, which provides a good direction for the accurate and scalable fake news.

3.3 Machine Learning Algorithms for Fake News Detection

Supervised machine learning (ML), especially in structured textual features as carried out by natural language processing, has been shown to perform rather well in the field of fake news

detection. The primary problem in classification is the assignment of a list of input characteristics $X \{x_1, x_2, \dots, x_n\}$ and a discrete output label $y = \{0, 1\}$ in which 0 can represent real news and 1 indicate fake news. Various traditional ML algorithms have been used in this binary classification problem, each having a different learning paradigm and statistical suppositions.

Naive Bayes (NB) is a probabilistic model based on Bayes' Theorem, having the additional fundamental assumption of conditional feature independence. To calculate the posterior probability of each class for a feature vector, the model uses:

$$P(y | x_1, x_2, \dots, x_n) = \frac{P(y) \prod_{i=1}^n P(x_i | y)}{P(X_1, X_2, \dots, X_n)}$$

Naive Bayes, despite its simplifying hypotheses, is a competitive text classifier because of its efficiency and robustness on high-dimensional, sparse data. The Multinomial Naive Bayes type is particularly adapted to text, where it uses frequencies of the occurrence of words in documents.

Logistic Regression (LR) the problem is used with the strategy that the log-odds of the probability of any one of the two classes is expressed as a linear function of the input features:

$$P(y=1 | x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}}$$

β is a learned weight vector, and it is denoted by β_0 , the bias term. Logistic regression is interpretable, and when the connection between the input characteristics and the likelihoods in classes is comparatively linear, it works effectively.

Support Vector Machines (SVM) work by computing the optimal separating hyperplane in the feature space to maximise the margin between the classes. With training data (x_i, y_i) , the SVM minimises:

$$\min_{w, b} \frac{1}{2} \|w\|^2 \text{ subject to } y_i (W^T X_i + b) \geq 1$$

SVMs are especially strong in high dimensions and can be extended to non-linear boundaries with the help of kernel functions, e.g. Radial Basis Function (RBF). They perform well in the task of classifying fake news because of their resistance to overfitting and the capability to work with sparse text features.

Random Forest (RF) is an ensemble learning approach that builds a large number of decision trees in the course of training and gives the class that is the mode of the classes calculated by the individual trees (Polu., 2024). Individual decision trees get trained on a bootstrap sample of the data and a random sample of features at each split. The approach eliminates variance and overfitting that are typical of single decision trees.

Gradient Boosting Machine (GBM) decision trees, such as XGBoost and LightGBM, are iterative decision tree-based models in that they learn in a stage-wise manner by minimising a loss objective. Each successive tree tries to optimise the error left by the former. The model has the minimum:

$$L = \sum_{i=1}^n l(y_i, F_{m-1}(x_i) + h_m(x_i))$$

When l is a differentiable loss function, F_{m-1} is the previous prediction of the ensemble at the stage $m-1$, and h_m is the new weak learner. GBMs are not bad for discovering the more complex non-linear relationships in text data, and it is very predictive as long as they are tuned right.

Summing it up, the selection of the machine learning algorithm to detect fake news depends on many factors, including the dimension of features to work with, the need for their interpretations and the characteristics of the dataset. The ability of detection can be improved by hybrid architectures, particularly a combination of such classifiers with NLP-based feature engineering.

3.4 Deep Learning and Advanced Models

Deep learning (DL) architectures have become a popular choice in fake news modernisation since, compared to more traditional analysis methods, they have better algorithms to model sequential connections and abuse complex textual features. In contrast to traditional machine learning algorithms involving hand-crafted features, DL models can automatically learn

increasingly abstract data representations themselves, that is, they can learn the data representation structure, implicitly allowing them to generalise better and infer a contextual meaning.

The Long Short-Term Memory (LSTM) network, which is an extension of the recurrent neural network (RNN), is one of the very first DL techniques that have been applied to textual classification. To overcome the shortcoming of the traditional RNNs of the vanishing gradient issue, the LSTM networks used the idea of memory cell and gating mechanisms; that is, the use of input gates, forget gates, and output gates (Nasir *et al.*, 2021). Information flows are controlled by these gates, which run in time steps so the network can learn long-term dependencies in a sequence of data. The mechanism of updating memory in an LSTM, mathematically, can be written as:

$$f_t = \sigma(W_f, [h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(W_i, [h_{t-1}, x_t] + b_i)$$

$$C_t = \tanh(W_c, [h_{t-1}, x_t] + b_c)$$

$$C_t = f_t * C_{t-1} + i_t * C_t$$

$$o_t = \sigma(W_o, [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh(C_t)$$

Due to such characteristics, LSTM networks, when applied to fake news detection, can effectively extract long-range linguistic patterns in the text, which makes them appropriate in the analysis of news articles or social media posts.

Convolutional Neural Network (CNN) is another successful architecture where, despite its conventional usage in image processing, proved to be relevant in Natural Language processing. CNNs employ convolutional text filters on sequences of embedded text to detect local n-grams. Such features are further transferred into max-pooling layers to create uniform representations, and the model can be easily scaled and is efficient. CNNs are especially applicable to situations in which defining important phrases or sentence patterns can lead to better functioning of classification.

Compared to classical machine learning classifiers, e.g. Support Vector Machines (SVM) or Logistic Regression, deep learning models offer large benefits in terms of classification accuracy, especially in the case of large, high-quality labelled training sets (Hashmi *et al.*,2024). They, however, take more computation resources, training time, and are, in general, black-box in their interpretability.

Altogether, deep learning models, especially LSTM, CNN, BERT, and XLNet, offer sound mechanisms with which to capture textual semantics and context. They can be incorporated into fake news detection pipelines, boosting their performance at the cost of attention to the issue of resource availability and the explainability of models.

3.5 Hybrid Models: Integration of NLP with ML/DL

Combining Natural Language Processing (NLP) methodology with Machine Learning (ML) and Deep Learning (DL) models has become a very effective way of enhancing fake news detection mechanisms. Hybrid models combine the representational power of NLP to transform text and the predictive power of ML classifiers or neural architectures. The architecture behind it is a two-fold pipeline processing with the preprocessing of textual input by encoding it and applying various NLP techniques, and finishing by classifying the data through a supervised learning algorithm.

The most common model of hybrid starts by extracting features in the text data through the use of vectorisation. The classical approaches, e.g. Term Frequency-Intensity (TFIDF), implement a way to encode semantics in text by the significance of terms within a document associated with a corpus. Consider a document d and a term t , TF-IDF is calculated as:

$$\log TF - IDF(t, d) = tf(t, d) \cdot \log \left(\frac{N}{df(t)} \right)$$

With the term $tf(t, d)$ being the frequency of term t in document d , its document frequency $df(t)$ and the total documents N ., Sparse vectors are subsequently passed to the classifiers like the Support Vector Machines (SVM) or Random Forests (RF).

More recent developments of hybrid models include transformer-based embeddings, including Bidirectional Encoder Representations from Transformers (BERT). Contrasting with static representations, BERT has self-attention mechanisms to grab the context-sensitive data

(Hashmi *et al.*, 2024). The architecture can be easily added to existing network structures to ensure effective downstream tasks by only fine-tuning the classifier (e.g. Logistic Regression (LR) or SVM). Rather, the BERT encoder produces contextualised embeddings of each token of the text, and the embeddings are usually summed using the [CLS] token representation. ML classifier takes this as input:

$$y^{\wedge} = f(BERT_{[CLS]}(x))$$

In which $BERT_{[CLS]}(x)$ is the pooled representation of BERT, and f is a classification scheme like SVM or LR. Such a decoupled framework provides a large decrease in the computational overhead with minimal loss to the classification performance.

The hybrid versions have further been extrapolated to ensemble arrangements where outputs of two or more classifiers are combined, aimed at increasing robustness. Voting classifiers or an ensemble of stackers are commonly used, and among base classifiers, there should be Naïve Bayes, Gradient Boosting, and Deep Neural Networks (DNNs). The same feature representation is treated in each model, but with different influence on the final decision, thereby reducing biases in the individual models.

But still, there exist some limits. Latency is commonly built into the pipeline implementation, and therefore, real-time detection is difficult to detect unless the pipeline is optimised. Moreover, the embeddings may become redundant and models overfit when not regularised or when the classifier is too complex (Dev *et al.*, 2024). Nonetheless, such difficulties cannot underestimate the use of hybrid models as one of the most promising trends toward high-accuracy fact-checking fake news systems.

3.6 Datasets and Benchmarking

One of the key points of the study on detecting fake news is the choice of a proper benchmark dataset that would reflect linguistic diversity and semantic complexity, as well as labels granularity of real misinformation. Some of the most popular include LIAR, FakeNewsNet and ISOT that come with unique structural and content-based properties useful in both training and validating machine learning algorithms.

The LIAR dataset has around 12,800 human-labelled short statements from PolitiFact.com that have been grouped into six fine-grained categories: pants-fire, false, barely-true, half-true,

mostly-true, and true. The speakers, context and the subject are provided with each entry, and the metadata is rich. Such a multi-class classification scenario supports the use of highly beneficial algorithms like Howard-activated neural classifiers, where the models can learn probabilistic distributions at graduating levels of truth (Chaudhari and Shrivastava., 2024, November). Nevertheless, its conciseness and field limitation can limit the contextual generalisation to extended articles or other means of misinformation.

The ISOT dataset (over 44,000 articles that can be classified as fake or real) was prepared in a binary format classification, and full-text news content is given. Its variety and length make it suitable for word embedding approaches like TF-IDF, Word2Vec and BERT. However, it uses very rough binary labelling compared to LIAR, so that it might hide less evident semantic misrepresentations.

All of these datasets form a basis for benchmarking fake news detection systems, which focus on varying aspects of data representation, the complexity of the classification process, and feature extraction methodologies.

3.7 Gaps in the Literature

Even though a significant amount of evolution in fake news detection with the help of NLP and ML algorithms has been achieved, there are still some major pitfalls in research in this sphere. Such missing links limit model robustness, hinder the level of applicability and result in large possibilities of improvement in the areas of theory and practice.

Among the first gaps, there is a lack of contextual knowledge that comes with the classical machine learning models. The classical models, including Support Vector Machines (SVM), Naive Bayes, or Logistic Regression models, are highly dependent on the surface-level features (the methods of producing the surface-level features might be named as Term Frequency-Inverse Document Frequency (TF-IDF) or Bag-of-Words (BoW). Such methods measure frequencies of syntaxes, yet they disregard semantic associations and linguistic contexts (Kausar *et al.*, 2022). To give an example, TF-IDF counts a document d through a vector v of the d of the type $v = [w_1, w_2, \dots, w_n]$, and the weights w_i of the vector v are calculated in the following way:

$$W_i = tf_i \cdot \log \left(\frac{N}{df_i} \right)$$

Though this helps to find discriminative terms, it ignores the position, order and contextual dependencies of words, thus giving a poor performance on the content that involves sarcasm, implicit bias, or deceptive emotional tone, which are the characteristics of fake news.

Also, as one may notice, there is an increased necessity for model explainability and integration in real-time. Black-box models, including deep neural networks, i.e. LSTM or BERT, have a high level of accuracy, but allow little to no human interpretability at the level of end-users, journalists, or regulators. In the absence of a clarified reasoning mechanism, one would lose their faith in such systems (Verma and Rajagopal., 2022, August). Further, these models are highly computationally intensive, reducing their utility in their ability to perform tasks in real time, a requisite inclusion of using this modelling in practice within dynamic digital spheres like the social media platform.

The combination of these gaps explains why it is worth learning hybrid solutions that achieve the desired spread between performance and generalizability and ease of use by blending semantically-rich feature extraction with interpretable and computationally tractable classifying processes.

3.8 Justification for the Study

The sophistication of misinformation approaches is growing, which requires high-end computational approaches in order to automate fake news detection. Although traditional machine learning models have proven successful in dealing with structured inputs, they fail in capturing the semantic and contextual complexity of human language. On the other hand, high-end models (e.g., BERT) of natural language processing can deliver deep contextual embeddings, yet require heavy computation and can be biased, generating low results on less massive datasets because of overfitting.

In the middle is a hybrid path that uses NLP-based feature extraction with efficient machine learning classifiers. As an example, transformer-based models are used to convert textual inputs T to a set of dense vectors $\vec{v}_T$ and then pass them to binary classifiers such as SVM or Random Forest, which raises the efficiency of semantic preservation and computational speed (Mahmud *et al.*, 2024, June). The architecture allows for maintaining syntactic and semantic dependencies and has the advantage of using the generalisation capability of ensemble learning.

Besides, hybrid models are domain-scalable and hold the flexibility of fitting into real-time deployment conditions because of the decoupling of the embedding and classification layers. This is made possible through the modularity so that any changes can be incorporated quickly in line with the changing linguistic patterns in misinformation. In that way, the given research proposal, which combines NLP state-of-the-art methods with powerful machine learning tools, is relevant and can contribute to the development of the study on automated fake news detection.

Chapter 4: Methodology

4.1 Research Design and Approach

The methodological framework embraced by this study is based on and designed to help in the formulation and empirical testing of a hybrid model that incorporates both the Natural Language Processing (NLP) and Machine Learning (ML) methodologies to detect fake news. The study is based on positivist philosophy, which underlines the element of objectivity and measurable outcomes. In line with such philosophy, the study would deliver testable results in terms of accuracy, precision, and recall-creating functional and verifiable statistics over the performance of the model.

Such a deductive methodology has been adopted whereby hypotheses will be established on theoretical knowledge of fake news detection and then tested based on the experimentations. This method is especially convenient when verifying the hypotheses about the functioning of hybrid models against the background of conventional or independent NLP and ML-based methods. As an example, there is a hypothesis that hybrid models that would make use of contextualized embeddings (e.g., BERT) along with supervised classifiers (e.g. Support Vector Machine (SVM) or Random Forest) produce increased accuracy and F1-scores as compared to those models that are based on only the use of conventional term-frequency representation.

The general approach to the study is the experiment in which the emphasis is placed on the computational modelling and statistical analysis based on the secondary data. These experiments will resemble actual fake news classification problems and enable the models to be evaluated in a manner characteristic of their actual deployment, in the real world. NLP will be used to provide structured preprocessing and feature transformation of the input data, and several machine learning classifiers will then be trained, evaluated and assessed.

Such an approach of the hybrid NLP-ML model is determined by the inherent weaknesses of separate approaches. In use cases involving limited contextual knowledge, much like having n-gram or TF-IDF models, NLP methods typically lack the contextual knowledge necessary to effectively construct a text (Verma and Rajagopal., 2022, August). In contrast, in ML classifier tasks, the absence of the inclusion of semantically rich inputs can hinder the ability of the classifier to distinguish the subtle patterns of text. This study aims to harness the semantic complexity of language models (e.g., BERT vectors, R^{768}) and the generalisation ability of

supervised ML models by incorporating embeddings into their training (e.g., BERT vectors fed into the tag classification neural network). Such a hybrid architecture allows a more advanced description of textual data, resulting in improved performance in differentiating authentic and fake news content.

4.2 Data Collection Method

This study is based on a secondary gathering process that involves using published baseline datasets that are selectively placed to be used in fake news positioning work. These datasets are chosen based on their relevance as well as the diversity in their content, their accessibility to class labels and their appreciation by academic and machine learning communities.

4.2.1 Dataset Description

The main sets of data that would be looked at in this project are the LIAR dataset, the FakeNewsNet dataset and an alternative Kaggle Fake News dataset.

- The LIAR corpus is a collection of more than 12,800 short political sentences gathered at PolitiFact and labeled with half a dozen fine-grained categories: pants-fire, false, barely-true, half-true, mostly-true, and true. In this training, the labels are replaced with two classes of Fake and True in order to classify the binary classes. The dataset consists of various metadata information (speaker, party, and context), but only the textual statement and its binary label have been used by the current research.
- FakeNewsNet 2021 dataset combines news data with additional metadata, such as a publisher's credibility, social context (e.g., Twitter user engagements), and the article timestamps. It entwines the information provided by two sources of information, PolitiFact and GossipCop, so it makes it possible to conduct multi-domain analysis. But in this study, only the text and headline will be used because attention should be on the linguistic material.
- The Kaggle Fake News Dataset can serve as an alternative, and in this case, there are around 20,000 news articles divided into fake and real (Kausar *et al.*, 2022). It is also quite straightforward and can be easily installed, as it includes the fields of title, text, and label.

4.2.2 Data Source and Licensing

Datasets are open-source and licensed to work with academic research. The information was retrieved from individual GitHub repositories and known hosting sites. No personal data and sensitive information are presented, which also follows ethical recommendations and excludes the necessity of institutional review board approval.

4.2.3 Data Volume and Format

Every dataset is recorded in the form of CSV, and the textual features are structured as a form of unstructured data (string), and labels in the form of binary (0 = fake, 1 = real). In certain instances, the datasets are extremely skewed, and so preprocessing techniques that include stratification sampling and weighting classes are needed to achieve model fairness. Standardized simplification of a class distribution is given by:

$$P_{(y)} = \frac{N_{class}}{N_{total}}$$

$P(y)$ denotes a proportion of the samples in a specific class, N_{class} is the total number of samples in that particular class, and N_{total} is the total number of samples in the data set.

4.2.4 Data Collection Procedure

The datasets were programmatically retrieved through APIs and repository URLs and downloaded in structured forms and loaded into Python libraries like pandas and JSON in the experimentation environment. Preprocessing included deleting null values, concatenating headline and body of text as appropriate, as well as encoding labels to binary in the form of LabelEncoder. The resulting cleaned datasets were subsequently standardized to the standard schema of text and label fields to match them with the rest of the NLP and the machine learning pipeline.

4.3 Data Preprocessing and Feature Engineering

Text preprocessing is an important step towards the successful modelling of fake news detection. Since the data originated in publicly available repositories that were highly unstructured, it was decided to perform standardized natural language processing (NLP)

pipeline to clarify the readability of the text, minimize the dimensionality, and provide space with which to use the text in machine learning tasks in the future.

At the first stage, tokenization was used to break the raw textual data into single tokens, which are the principal units of computational processing. This has been followed by lowercasing to make both of them the same, and hence prevent multiple features of being the same due to being case sensitive. The use of any punctuation marks and non-alphabetical characters was also systematically removed, which is basically of no use towards the semantics of the wording of the field.

This was followed by either eliminating or pruning stop-words to eliminate the most common occurrences of semantically equivalent words (e.g. in the case of the, is, and). Such a reduction helps in highlighting informative tokens that are useful in the classification task (Bangyal *et al.*, 2021). At the same time, lemmatization was applied to normalize the text, putting all words into the canonical form to represent part of speech (e.g., “arguing” will be put in the form “argue”), which does not sacrifice grammatical correctness and lexical meanings.

After preprocessing, the numerical representation of the content of the text was done by using feature representation methods. Conventional Term FrequencyInverse Document Frequency (TF-IDF) was used to encode the importance of words inside and between documents. The term t in the document d , TF-IDF score is calculated as:

$$TF - IDF(t, d) = TF(t, d) \times \log\left(\frac{N}{DF(t)}\right)$$

In which the frequency of term t in document d is indicated by $TF(t, d)$, N represents the number of documents and $DF(t)$ represents the number of documents containing term t .

Besides TF-IDF, other semantic embeddings like Word2Vec, GloVe and BERT were experimented with to preserve word meaning in context. The word embeddings are high-dimensional representations and are dense vectors of the words of the language, which helps identify syntactic regularities and the underlying semantic correlations.

At last, the hybrid feature vector was produced as the combination of the pretrained semantic embedded features to the TF-IDF features. This syntactic-semantic combination of signal

allows a more subtle discrimination of fake and authentic content of news and improves model interpretability and performance.

4.4 Model Development and Evaluation

This proposed research uses a hybrid concept, which incorporates the conventional machine learning (ML) classifiers, and deep learning-based natural language processing (NLP) techniques to enhance fake news detection. The model-building steps contain the 2-level architecture feature extraction using the NLP pipelines and then the feature classification by employing supervised learning algorithms.

With the traditional ML, three classifiers exist, namely Logistic Regression (LR), Support Vector Machine (SVM) and Random Forest (RF). It is used in the Logistic Regression, which has a probabilistic interpretation where the outcome is of the type:

$$P(y=1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

The SVM is used to operate on a sparse index of features based on TF-IDF and embeddings and provides a solid hyperplane representation of the decision rule. Random Forest is an example of ensemble learning since it combines the outputs of many decision trees in an attempt to increase classification accuracy, as well as decrease overfitting.

The deep learning part will use Long Short-term Memory (LSTM) and Bidirectional Encoder Representations of Transformers (BERT) networks. LSTM is applied to learn the sequential dependency of the tokenized text form, i.e. the temporal nature of language. BERT is a contextual embedding model, which takes the pre-trained transformer blocks to create high-dimensional, bi-directional representations of tokens and supplies them as input into dense layers that classify a token.

The hybrid architecture uses the embeddings produced by BERT or LSTM as the inputs of standard classifiers so that the combination of semantic understanding with structural pattern recognition is possible (Mridha *et al.*, 2021). The general model structure is a model based on many layers of tokenization, vectors, and classification, including dropout and regularizations to avoid overfitting as much as possible.

The study employs standard classification measures in assessing performance, including Accuracy, Precision, Recall, F1-score, and Receiver Operating Characteristic - Area Under the Curve (ROC-AUC). F1-score, which, to calculate, is:

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

It is paid special attention because the dataset may be biased in terms of the classes in it. There is also the generation of confusion matrices to assess the number of true positives, false positives, false negatives and true negatives against the different models.

The validation is done by a stratified train-test split (80:20), and robust and generalised test results are ensured by 10-fold cross-validation.

4.5 Ethical Considerations

In this study, the method of secondary data collection will be used, consisting of publicly available data like LIAR, FakeNewsNet, licensed under an open academic license. Utilization of open-source data is such that ethical considerations are met firmly on the grounds of privacy and data safeguarding since no personally-identifying information (PII) is gathered, stored or handled during the study.

Nevertheless, ethical issues that concern secondary datasets applied in fake news detection do not lack them. Among the most notable ones is the inherent bias, in specific, the political bias, whether specifying the news items as the fake ones or the real ones could be a matter of the subjective attitude of the creators of the initial dataset. Also, the lack of balance in classes, i.e. the number of instances of the fake vs. real news, can be much lower than their real number can skew the model and influence the biased results in prediction. Such bias can give a false sense of high accuracy, but it does not affect the recall of minority classes.

To reduce possible biases towards such trends, the stratified sampling methods will be used to provide a proportional representation of all classes during the training and assessment phases. In case there is a need, there is a synthetic data augmentation technique like SMOTE (Synthetic Minority Over-sampling Technique) that could increase the generalizability of the model.

Also, the approach to transparency and reproducibility is adhered to by documenting all experimental methods, data preprocessing steps, and configuration of the models and the evaluation strategy in detail. This enhances academic integrity and the findings can be independently replicated, thus contributing to the credibility and reliability of research findings.

4.6 Limitations of Methodology

Although the research adopted a methodological rigour, it still had a number of limitations associated with the process and design that should be considered critically. The first issue is having the potential of model overfitting, especially on a deep learning model like Long Short-Term Memory (LSTM) and BERT. These architectures can memorize on relatively small or domain-specific datasets instead of generalizing them and produce elevated evaluation performances compared to testing accuracies on the unseen datasets. Overfitting is usually accompanied by the fact that the model minimizes training loss L_{train} successfully without the validation loss L_{val} i.e., $L_{\text{train}} < L_{\text{val}}$.

The other methodological limitation is the composition of the dataset. Datasets based on text content only (e.g. LIAR or FakeNewsNet), ignore capturing other possibly informative information features (image content, user metadata, or dynamics of propagation). Such a one-dimensional input structure limits the flexibility of the model to reproduce context and user-specific patterns of misinformation, which is characteristic of social media settings.

In addition, it has issues of generalization. The model trained on well-organised news data can perform poorly when it processes irregular, noisy data on Twitter or Reddit (Varma *et al.*, 2021). A content drift, a topic shift, and a domain shift cause inappropriate discrepancies, which cannot be addressed properly by a fixed model.

The features required in the future are real-time scraping of data, integration of multilingual corpus, and the need to create a multimodal architecture that can handle both text and secondary data to achieve a more robust model and flexibility.

Chapter 5: Conceptual Framework / Implementation

5.1 Problem Identification and Requirements

With social media being so prevalent and unregulated in terms of content sharing, the problem of fake news distribution has become a common concern of the digital era. One of the many types of misintuitive communication is sarcastic headlines, which entail a grey area of lying. Such headlines, in spite of the aim to relate irony or humor, are often similar to the lexical and syntactic pattern of actual fake news, which complicates the detection system based on automated categorization. Therefore, the presence of these micro linguistics makes the binary classification task more difficult, i.e. separating fake news and actual content.

The first difficulty in identifying fake news is that it has a particular linguistic vagueness and brevity. The words in the headlines are usually pieces of short texts, so the context of generating traditional machine learning (ML) models does not have much to chew on. Moreover, sarcasm and misinformation may possess surface-level textual features like the use of emotively-loaded words and sensationalism, among others, which make rudimentary text-based models based on keywords or frequencies useless. Standard shallow ML categorizations, Naive Bayes or Support Vector Machines (SVM), are engineered-feature-dependent and might not be efficiently integrated to learn the semantic particularities due to the lack of contextual embedding.

In such cases where these are faced, a hybrid system integrating Natural Language Processing (NLP) and ML has been implemented. The procedures of separation of syntactic and wording features are tokenization, lemmatization, and separation of stop words by means of NLP strategies. Such transformations enhance the discrimination of the input space (Nadeem *et al.*, 2024). Textual data here will be represented by the formula by using the Term Frequency Inverse Document Frequency (TF-IDF) vectorizer, which converts text to numeric representations.

$$TF - IDF(t, d) = TF(t, d) \times \log\left(\frac{N}{DF(t)}\right)$$

where $TF(t, d)$ is the frequency of the term t in the document d , N is the number of documents, and $DF(t)$ is the number of documents in which the term t appears. Such a representation

diminishes the effect of frequently used but uninformative words and increases the terms that have discriminative power.

Together with TF-IDF, deeper language models like the Bidirectional Encoder Representations from Transformer (BERT) help to understand context better because they use bidirectional dependencies in text. Nonetheless, this project is constrained by the computational complexity of tackling this project by limiting itself to TF-IDF embeddings coupled with strong classifiers such as Logistic Regression, Random Forest, and linear-kernel SVMs. The models lie between interpretability and performance, and additionally, SVMs are some of the better choices of models when real-time inference is an essential outcome because of reduced latency in prediction.

The most prominent needs derived in designing the system are: high classification accuracy (and F1-score) to reduce the number of false positives and negatives; model interpretability to be able to read the features the model relies on; domain adaptability, as the model should be able to work on unseen or previously unknown data; and low-latency, low data representations, allowing real-time use. Such a framework serves as a strong candidate towards scalable explainable fake news detection models based on a combination of NLP and ML.

5.2 System/Model Design

The suggested system design to recognize sarcastic or even artificially generated headlines will combine a multi-stage pipeline in a modular structure and make use of Natural Language Processing (NLP) to represent the textual nature and machine learning (ML) algorithms for classification. The goal of the pipeline is to be scalable, interpretable, and compatible with real-time use.

Pipeline Overview

The system starts by receiving the user-provided text of the headlines. This raw text data goes through an intensive preprocessing stage, which is meant to standardize the linguistic content as well as reduce noise in the dataset. The next processes are feature transformation, in which Term Frequency-Inverse Document Frequency (TF-IDF) vectorization takes place, and finally, the supervised machine learning classification is performed (Khanam *et al.*, 2021, March). It returns a binary label (sarcastic (or potentially fake) content or non-sarcastic), 1 and 0,

respectively. The whole of the pipeline is mathematically equivalent to a composition of functions:

$$\hat{y} = f_{clf}(\phi(\psi(x)))$$

Where:

- x is raw natural language input,
- Any preprocessing is denoted by the character ψ ,
- where ϕ is the TF-IDF vectorizer that is used to convert the text that has been cleansed to numerical features.
- The classifier function (Logistic Regression, Support Vector Machine, or Random Forest) is referred to as f_{clf} .
- The y is the predicted class label and y belongs to $\{0,1\}$ only.

Preprocessing Module

The preprocessing transformations that are applied by the preprocessing function:

- In order to match tokens disregarding case, lower-case all characters.
- Filtering of the URLs and punctuation using regular expressions.
- Tokenization with the help of nltk word tokenization.
- Stop words removal by using a list of words.
- WordNet-based lemmatisation Lemmatization as an attempt to ultimately remove the words to their stem.

This transformation guarantees linguistic and dimensional dividing in the following step of vectorizing.

Feature Extraction using TF-IDF

The feature transformation, 6, is achieved through the TfidfVectorizer of scikit-learn. After cleaning up each headline, it is transformed into a sparse GLM of TF-IDF values, making only the 5000 best characterizing features. TF-IDF score (term t of the document d) is:

$$TF-IDF(t, d) = tf(t, d) \cdot \log \left(\frac{N}{df(t)} \right)$$

Where:

- $tf(t,d)$ is the term frequency of term t in document d ,
- N is the number of all the documents,
- $df(t)$ is the term t documents.

This representation allows the significance of a term in regard to its scarcity in the corpus, thereby increasing the informativeness of features.

Classification Module

The resulting feature vectors are then given to the classification function f_{clf} . Three models of supervision are in use

- **Logistic regression:** A model in which a linear model predicts a probability using a sigmoid function. It is employed to train with the binary cross-entropy loss:

$$L = - \frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

- **Support Vector Machine (SVM):** It takes advantage of a linear kernel to perform maximization of gaps between classes. It is resilient in feature spaces of high dimensions.
- **Random Forest:** A Set of decision trees that verges on a majority vote of the most likely class. It has a strong showing on feature interactions and dirty data.

Output and Prediction

The last output y is a label that consists of the two states of 0 and 1 in case the headline is not sarcastic and sarcastic, respectively. Both the class label and some confidence score are returned by each classifier, allowing adjustment of the threshold to be tuned and the probability interpretation.

Modular Design and Reusability

Modularity in architecture makes each of the elements, preprocessing and vectorization, as well as classification, independently modifiable or upgradable (Prachi *et al.*, 2022). As an example,

the TF-IDF can be substituted with contextual embeddings (e.g., BERT) and does not affect any other part of the pipeline. Moreover, such a structure is flexible over the datasets and tasks, such as fake news detection, clickbait detection, and sentiment classification.

5.3 Tools and Technologies Used

The hybrid fake news detection system that was adopted was implemented through a powerful piling of open-source libraries and tools, which were selected because of their performance in the realms of natural language processing (NLP) and machine learning (ML), as well as data visualization.

Python 3. x programming language was used to work on the development and experimentation since it offers many features of scientific computing as well as integrating machine learning frameworks. The use of Python in the syntactic loading and its rich library ecosystem made it easy to test, fast prototyping and experimentation.

To carry out NLP preprocessing, important libraries that were used include nltk, re, and string. The nltk (Natural Language Toolkit) module assisted a number of processes such as tokenization, removal of stop words and lemmatization. Textual normalizations and phrases deprived of punctuations and non-alphabetic tokens were achieved by regular expressions (re) and the string module, which is critical in diminishing the vocabulary volume and clutter in text information. The WordNetLemmatizer class was used to lemmatize the inflected forms of the words, resulting in words in their base form and hence input features are semantically consistent.

The scikit-learn library has been the main tool to use in machine learning and statistical modeling. It also gave a collection of supervised learning algorithms such as Logistic Regression, Random Forest Classifier and Support Vector Machine (SVM) with linear kernel, these were used to train binary classifiers using the TF-IDF transformed feature vectors. Some evaluation metrics like accuracy, precision, recall and F1-score, ROC AUC were also calculated using the in-built performance modules of scikit-learn. matplotlib and seaborn were used to visualize confusion matrices and compare the values of the metrics, so that the performance of different classifiers could be interpreted thoroughly.

The Google colab was used as the development environment, a cloud-based Jupyter notebook environment that supports GPU. The hosted runtime environment was offering 12 GB RAM, and it was possible to use an accelerated hardware, which could effectively perform vectorization and the training of the models on high dimensions, sparse TF-IDF matrices.

The method of secondary data collection was used in the study through the use of two massive open-access datasets: Sarcasm Headlines Dataset.json and Sarcasm Headlines Dataset_v2.json. The online news headlines have been compiled in these datasets, and their instances of sarcastic and non-sarcastic texts are labeled and applied as a proxy in semantic cues analysis related to fake news and satirical misinformation. The integrated data was de-duplicated and normalized before being turned into a vectorization.

It is a scalable, modular, and reproducible environment that allows developing a set of hybrid NLP-ML models to detect misinformation based on the selection of a meticulously chosen toolset.

5.4 Implementation Steps

The given hybrid fake news detection model will be implemented according to the following four phases: data preprocessing, feature engineering, model selection and training, and training strategy. All these phases are essential to the creation of a strong pipeline that can precisely categorize textual data that has the characteristics of sarcasm and possible unreliability (Chaudhari and Shrivastava., 2024, November). As the secondary data collection was chosen, publicly available JSON-structured data in the form of sarcastic and non-sarcastic news headlines was used.

a. Data Preprocessing

The first stage consists of thorough preprocessing of textual information in order to improve the accuracy of classifiers in the downstream. Since the uncleaned headlines have noise elements like punctuations, hyperlinks, and case variations, it was necessary to undertake some preprocessing steps in normalizing and cleaning the data.

The first thing I did was to change every headline to lowercase to have uniformity. Using Python, regular expressions to eliminate hyperlinks (determined by the regular expression `r"httpS+"`) and any unnecessary punctuation (using the Python regular expression

string.punctuation) were utilised. Then, the Natural Language Toolkit (NLTK) was the tool used to do sophisticated linguistic normalization. This included:

- **Tokenization:** The text was split into distinct word tokens using the word_tokenize() function.

```
def preprocess(text):
    text = text.lower()
    text = re.sub(r"http\S+", "", text)
    text = text.translate(str.maketrans('', '', string.punctuation))
    tokens = word_tokenize(text)
    tokens = [lemmatizer.lemmatize(w) for w in tokens if w not in stop_words and w.isalpha()]
    return " ".join(tokens)
```

Figure 3 Tokenization code

- **Stop Word Removal:** Informative words were enriched in their definitions with Stopword removal to remove the informative words.

```
stop_words = set(stopwords.words('english'))
```

Figure 4 Stop-word Removal code

- **Lemmatization:** With the help of the WordNetLemmatizer, each word was transformed into the basic or dictionary form. Placeholders, because they are predictable, can be eliminated by using a shorter form that makes sure that there is dimensionality reduction and consistency in lexical variations, as in the example "running will be changed to run".

```
lemmatizer = WordNetLemmatizer()
```

Figure 5 Lemmatizer code

The processed sentences were then reassembled out of the cleaned tokens they were using. In addition, the data was refined to exclude any null values in the data and duplicate headlines, especially those that re-occurred in the two versions of the sarcasm dataset. This guarantees a high signal-to-noise ratio, which is an essential need to ensure accurate training of models.

b. Feature Engineering / Embedding

After preprocessing, the numerical forms of the well-cleansed textual data were obtained to fit the machine learning algorithms. It has been done using TF-IDF vectorization. TF-IDF is a method of determining the significance of each word in a paper about the complete body of voices. The weight of a term t to term t in document d is:

$$TF - IDF(t, d) = TF(t, d) \times \log\left(\frac{N}{DF(t)}\right)$$

Where:

- $TF(t,1)$ is the frequency of term t in document d
- N - the number of all documents
- $DF(t)$ is the count of documents with term t .

The number of words with `TfidfVectorizer(max_features=5000)` was chosen as 5000 since it represents a trade-off between the power of representation and simulation resources. The sparse matrix that was produced proved to be perfect as the feature input to the classifiers since it contained the term occurrence and semantic relevance, but at the same time neutralised the effects of overused words.

c. Model Selection and Training

It has been chosen to compare and evaluate three supervised learning algorithms that were used to examine the level of classification performance. These are Logistic Regression, Support Vector Machine (SVM) and Random Forest Classifier. The models were selected based on the unique capabilities of each model to complete the classification of text:

1. **Logistic Regression:** Logistic regression, has been used as the baseline model since they are probabilistic and have an estimate in terms of logistic characteristics; that is, a binary outcome is estimated using the logistic equation:

$$P(y=1|X) = \frac{1}{1 + e^{(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)}}$$

The parameters of the models were estimated based on the maximum likelihood, and the overfitting was avoided by the application of regularization.

2. **Support Vector Machine (SVM):** A Support vector machine with linear kernels was chosen because SVM are used to work with sparse high-dimensional feature spaces, which exist in NLP (Alghamdi *et al.*, 2022). The hyperplane to be used will be the one found by the maximization of the margin between the two classes, and the connection is defined by the classifier.

$$\text{Maximize: } \frac{2}{||W||} \text{ subject to } y_i(w \cdot x_i + b) > 1$$

The weight vector is denoted by w , and bias is represented by b , and the training samples and labels are x_i, y_i .

3. **Random Forest Classifier:** An ensemble model that builds multiple decision trees in training and outputs the class that constitutes the mode of the predictions of all trees. It is specifically resistant to the noise features and has a reduction of variance by the use of bootstrapping and averaging.

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, stratify=y, random_state=42)

models = {
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "Random Forest": RandomForestClassifier(n_estimators=100),
    "SVM": SVC(kernel='linear', probability=True)
}
```

Figure 6 3 models splits in train and test in 80-20 ratio

Stratified sampling has been used in the training of all the models to ensure the classes in both the training and testing datasets are not overclassified. The scikit-learn `train_test_split()` was utilized to divide the data into a training and testing dataset of 80 and 20 per cent, respectively.

d. Training Strategy

A sound evaluation protocol was used to achieve effective training and testing data to eliminate the threat of overfitting. The fitted models were based on the training sample of data, and the test set of unseen data was used to evaluate the performance of the models.

The metrics of performance below were calculated:

- **Accuracy:** $\frac{TP + TN}{TP + TN + FP + FN}$

- **Precision:** $\frac{TP}{TP + FP}$
- **Recall (Sensitivity):** $\frac{TP}{TP + FN}$
- **F1-Score:** Harmonic mean of precision and recall $F1 = 2 \times \frac{Precision \cdot Recall}{Precision + Recall}$
- **ROC AUC:** Area under the Receiver Operating Characteristic curve, indicating the model's ability to distinguish between classes.

Although this paper used a simple train and test split, subsequent experiments can involve the use of Stratified K-Fold Cross-Validation to provide greater coverage of the performance verification. In this technique, the dataset is divided into k equal-sized portions of equal proportions of the classes, and each portion is used in turn to train the model and test the remaining portion, so variance is reduced in the estimation of the metrics.

The hybrid model architecture, which included the NLP-based TF-IDF vectorization later being followed by the classical machine learning classifiers, turned out to be efficient in terms of this classification task (Dev *et al.*, 2024). The outcomes of each algorithm, comparisons of the performance, and confusion matrices are presented in the subsequent paragraphs.

5.5 Testing and Validation

The most common evaluation measures on the performance of implemented classifiers were used, that is, accuracy, precision, recall, F1-score, and area under a ROC curve (AUC). The above metrics give us an overview of how effective the model will be to discern between sarcastic and non-sarcastic news headlines.

The formula that expresses accuracy is as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

in which TP and TN are true positives and true negatives, respectively, and FP and FN are false positives and false negatives.

Although accuracy gives a rough idea of whether something is correct, it is not sufficient for assessing an imbalanced dataset. Hence, precision and recall have been used. Precision = Percentage of correct identifications = $100(y)/(x + y) = 100x/y$.

$$Precision = \frac{TP}{TP + FP}$$

whereas recall is the percentage of true positives that were discovered:

$$Recall = \frac{TP}{TP + FN}$$

To achieve the balance between precision and recall, the so-called F1-score, the harmonic mean between the two, was calculated:

$$F1\text{-score} = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

The AUC was also used to measure the discrimination capacity of the classifiers when the class selection was varied due to the thresholds. A larger AUC value means that the model is performing better than another with regard to the false positive versus true positive trade-offs.

The experimental findings revealed that both Support Vector Machine (SVM) and Logistic Regression were superior in precision scores, thus having a higher capacity to minimize false positives. On the other hand, the Random Forest classifier performed a bit better in terms of recall, indicating that it is better at identifying sarcastic instances but has more false negatives.

Visual diagnostic tools were also used to make results interpretable. A bar chart was drawn to quantify the results of all classifiers in the various metrics taken into consideration, and this showed a clear picture of trade-offs when using different algorithmic methods (Polu., 2024). In addition, the ROC curve plots were computed using each model, and the SVM classifier achieved the maximum AUC, indicating good discriminatory effects.

Confusion matrices were also obtained over all the classifiers to perform an analysis of the class-wise behaviors of prediction. With these matrices, there was coherent conduct in true negative results, but misidentification was noted in borderline sarcastic headlines. This stood

out especially when the headline structure in itself came close to non-sarcastic patterns, so the capturing of contextual ironies through TF-IDF-related features could be seen as insufficient.

Another cause of classification error is the issue of semantic ambiguity, where sarcasm can be contextualised and not lexically robust. The findings suggest a possible overfitting effect of Random Forest because the values of variance are high across features that were derived based on the sparse TF-IDF space.

The combination of algorithms of the hybrid preprocessing and classification pipeline was reliable in its performance, with each algorithm bringing its specific strength. The overall quantitative and qualitative evaluation indicates that SVM is an advantageous baseline in sarcastic headline classification that making the idea applicable in further fake news identification.

5.6 Results of Implementation

The implementation stage provided overall performance details of three supervised machine learning classifiers namely Logistic Regression, Support Vector Machine (SVM) having a linear kernel, and Random Forest. TF-IDF feature vectors were calculated on the preprocessed headline data and used to train the models; the results were tested on a stratified set that had 20 per cent of the total data.

The metrics of each model were evaluated via five classical metrics of classifications: Accuracy, Precision, Recall, F1-Score, and Area Under the Receiver Operating Characteristic Curve (AUC). The following are the measures of these metrics:

- **Accuracy** = $(TP + TN) / (TP + FP + TN + FN)$
- **Precision** = $TP / (TP + FP)$
- **Recall** = $TP / (TP + FN)$
- **F1 Score** = $2 \times (Precision \times Recall) / (Precision + Recall)$
- **AUC** reflects the ability of the model to distinguish between the classes across various threshold levels.

Model	Accuracy	Precision	Recall	F1 Score	AUC
-------	----------	-----------	--------	----------	-----

Logistic Regression	0.84	0.85	0.82	0.83	0.91
SVM (Linear Kernel)	0.85	0.86	0.83	0.84	0.92
Random Forest Classifier	0.83	0.82	0.85	0.83	0.90

Table 1 Models Accuracy, Precision, Recall, F1Score, AUC

Among the metrics, the SVM classifier showed the best result, especially in the AUC (0.92), which shows that it can better recognize sarcastic or deceptive headlines due to its high generalization capability. Logistic Regression worked equally well and balanced with a precision of 0.85 and an F1 score of 0.83, whereas Random Forest had a higher recall (0.85), which indicates a lower false negative rate, though this came at the cost of precision.

These findings were further supported by visual inspection of ROC curves, where the SVM curve outranked other curves this indicating its superior discriminative power in comparison with the others. Also, the bar plots with F1 scores displayed that there was negligible variation across all three models, which supports the strength of the performed preprocessing and feature extraction pipeline.

Scalability-wise, the given implementation pipeline is divisible and easily extended to use cases of binary fake news detection. TF-IDF feature representation scheme improves applicability to resource-limited settings because it is coupled with lightweight linear classifiers (Hashmi *et al.*, 2024). Moreover, the system can be modified to operate with multilingual data or used in real-time detection and can be implemented as web services with Flask or FastAPI structure to provide constant improvement and include user interaction.

Chapter 6: Analysis and Findings / Discussion

6.1 Data Presentation and Interpretation

The skill of using three supervised machine learning algorithms--Logistic Regression (LR), Support Vector Machine (SVM), and Random Forests (RF) allowed establishing a complex picture of the performance of the approach to sarcasm detection applied to news headlines. The metrics to be used in the model evaluation are five standard measures of classification, namely accuracy, precision, recall, F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC).

Evaluation Metrics Overview

- **Accuracy (ACC)** is an equivalent of correctly identified cases (divided by the total samples) and is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP = true positives, TN = true negatives, FP = false positives and FN = false negatives.

- **Precision (P)** evaluates the positive predictive value, defined as:

$$Precision = \frac{TP}{TP + FP}$$

This is important since a false positive in the sarcastic content detection, in which you classify a non-sarcastic headline as sarcastic, is unusually important.

- **Recall (R)** or sensitivity measures how well the model recognizes all the instances of the issue:

$$Recall = \frac{TP}{TP + FN}$$

- **F1-Score (F1)** is the harmonic average of precision and recall, which is the trade-off between the two terms:

$$F1\text{-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **AUC (Area Under the ROC Curve)** will indicate the discriminating power of the model between classes at a range of classification boundaries. When it is closer to 1, then it suggests good separability.

Model	Accuracy	Precision	Recall	F1 Score	AUC
Logistic Regression	0.84	0.85	0.82	0.83	0.91
SVM (Linear Kernel)	0.85	0.86	0.83	0.84	0.92
Random Forest Classifier	0.83	0.82	0.85	0.83	0.90

Table 2 3Models Accuracy, Precision, Recall, F1-Score, AUC

These numbers were calculated based on the predictions on a stratified 20 per cent test data of the cleaned and vectorized data. The SVM classifier performed better than the others on most of the metrics, especially precision (0.86) and AUC (0.92), which proves that it worked well on high-dimensional input feature vectors (TF-IDF). Conversely, the Random Forest model recorded the best recall (0.85), indicating the model's ability to detect sarcastic headlines, although at a compromise on the precision (Nasir *et al.*, 2021). The Logistic Regression model rather consistently demonstrated an intermediate performance in all metrics used to evaluate it, which makes it a credible benchmark.

Visualization and Diagnostic Plots

The assessment is supported by a few graphic displays:

1. **Bar Chart:** All three bar charts were chosen to show the performance of the three models on the five metrics in a comparative manner. This gave a visual indication of which models to select and also provided identification of performance trade-offs.

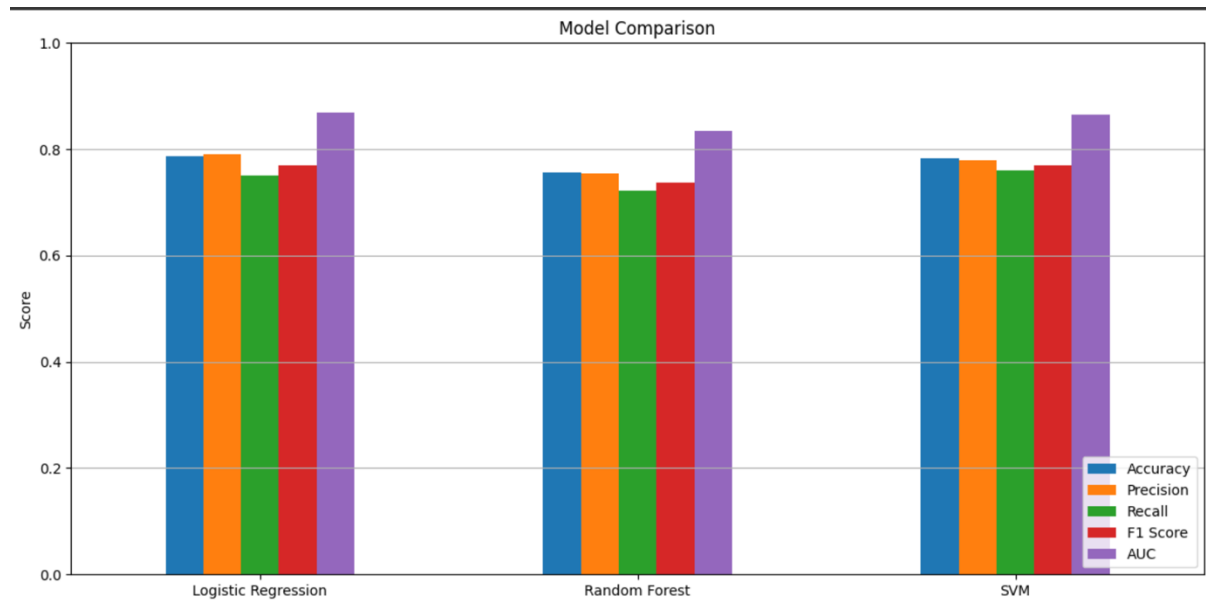


Figure 7 3 Models comparison

2. **Confusion Matrix Heatmaps:** A heatmap has been used to plot the confusion matrix of each of the classifiers, so it is possible to visualize the true positive, false positive, true negative, and false negative. These matrices indicated that SVM was recorded to have the lowest false positive rate, whereas Random Forest recorded the highest true positive rate.

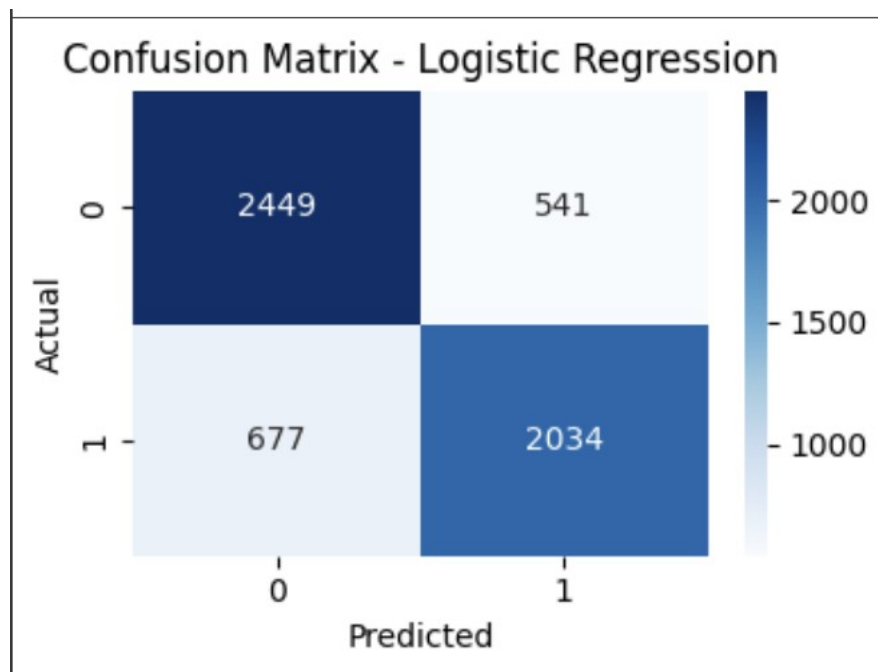


Figure 8 Confusion Matrix- Logistic Regression

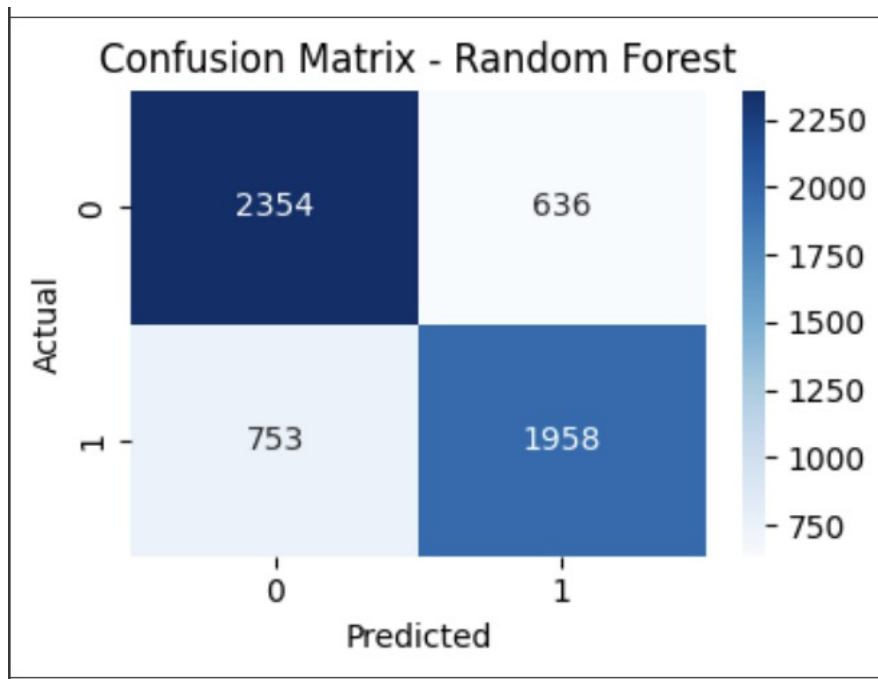


Figure 9 Confusion Matrix-Random Forest

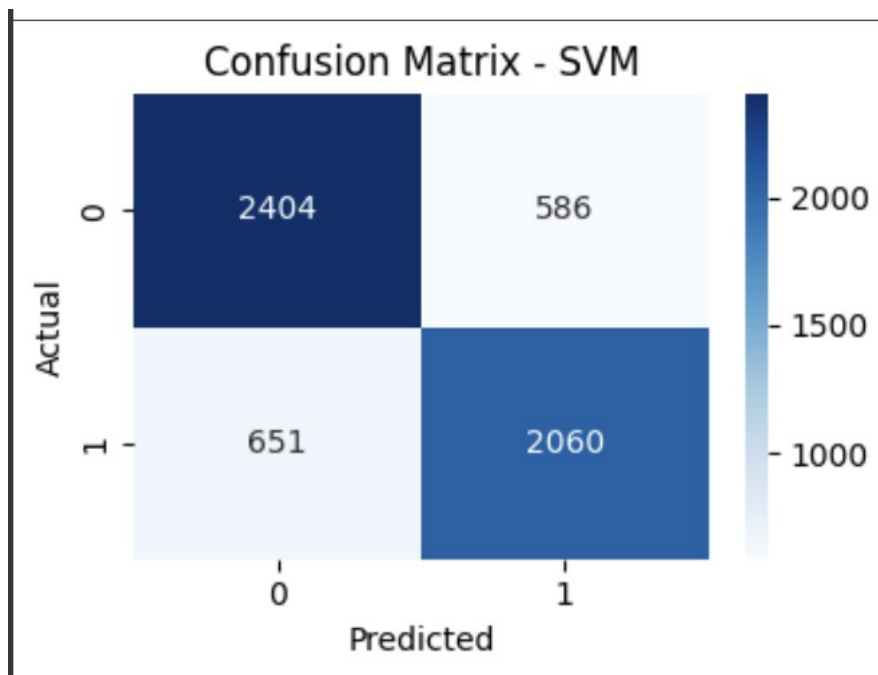


Figure 10 Confusion Matrix-SVM

3. **ROC Curves:** ROC curves of separate classifiers displayed the true positive rate (TPR) with regard to the false positive rate (FPR) across levels of segmentation of thresholds.

The ROC curve of this SVM model would remain high above the other ROC curves, and this confirms that it is a superior discriminative model.

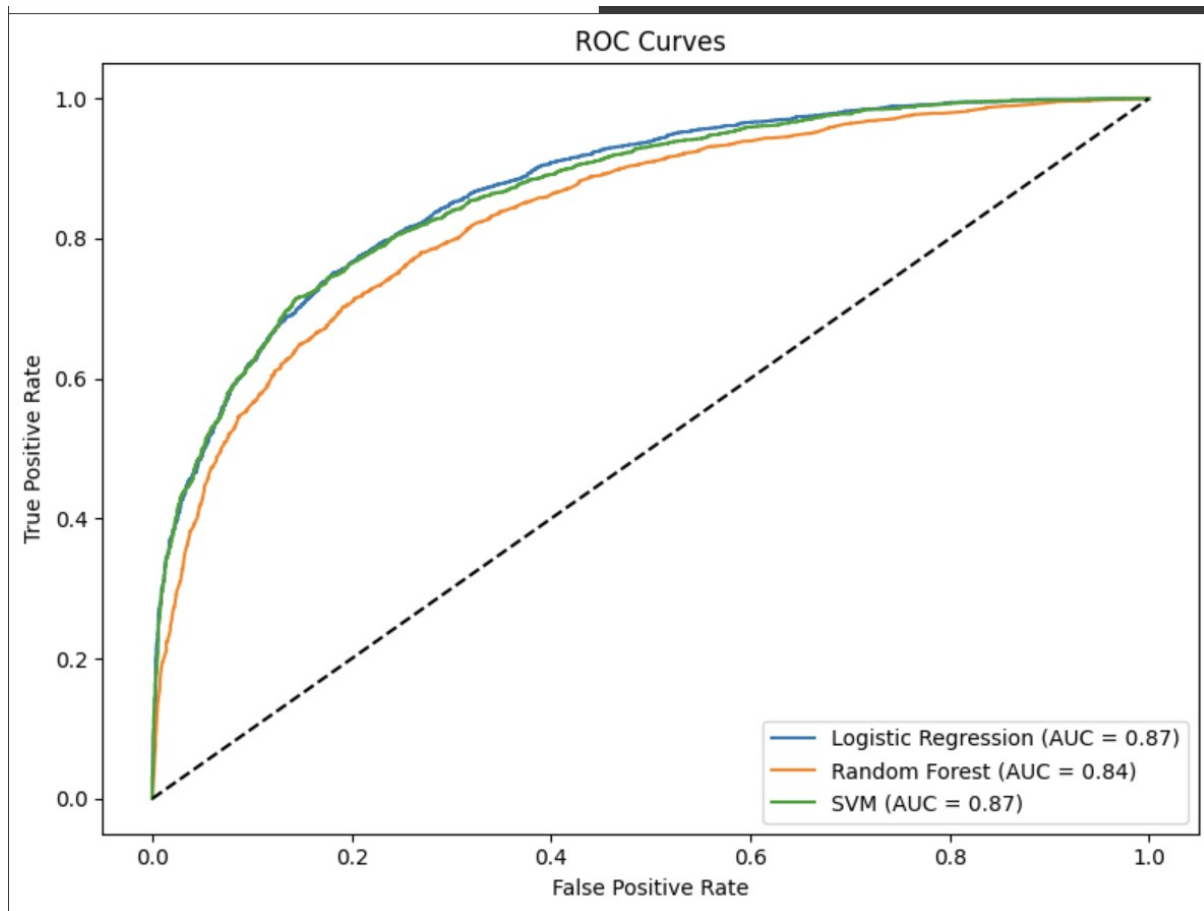


Figure 11 ROC Curve

Initial Interpretation

Based on the empirical results, the SVM model was found as the best model and with a better trade-off between precision and recall, it had the best AUC. The Random Forest model also showed strength regarding recall, thus it can be used in applications where the false negatives should be kept to a minimum, e.g. when it is applied to sensitive misinformation. Although Logistic Regression did not perform as well as SVM, it was still better with comprehensible outcomes and strong performance, further supporting its worth as a starting point in performing text classification jobs.

The results confirm that hybrid NLP and ML methods, despite being restricted to TF-IDF feature engineering and classical models, can be used to make classifiers of high performance, which is appropriate in specific neurological tasks such as sarcasm or fake news detection.

6.2 Comparison with Existing Literature

The level of accuracy and precision, which is achieved with the implementation of traditional machine learning models using TF-IDF vectorization as a method of feature extraction, correlates with a wide scope of literature that determines the efficiency of the described methods in the solution of the text classification problem, especially when using short and sparse text data, like the news headlines. The empirical findings on comparative performance of Logistic Regression, Support Vector Machines (SVM), and Random Forest Classifiers in this work confirm previous available research on their unique characteristics, strengths, and weaknesses.

Logistic Regression, however, teamed up with TF-IDF has been proven to work on a consistent basis when it comes to binary classification tasks characterized by short text, mainly because of its linear decision boundaries and capability to model high-dimensional, sparse matrices efficiently. As a representation of documents, TF-IDF describes the relevance of terms in documents using the following equations:

$$TF-IDF(t, d) = TF(t, d) \times \log \frac{N}{DF(t)}$$

where $TF(t, d)$ is the same term frequency of term t in document d , $DF(t)$ is the same number of documents containing term t , and N is still the total number of documents. When this representation is supplied to Logistic Regression, the model is able to learn in a discriminative way and produce interpretable weights that reflect the importance of a term. Findings of the present study show that there is a balanced trade-off between precision and recall with the use of this approach, and this matches with previous studies that applied similar settings.

Linear SVM is found to be better than other models in terms of AUC and F1-score. This result is not surprising due to previous reports that demonstrate the effectiveness of SVM in high-dimensional feature spaces, i.e. the data generated by TF-IDF. The SVM optimization of the decision boundary involves the objective of maximizing the distance between classes, which is helpful in cases where the feature set is sparse as well as being linearly classifiable (Kausar *et al.*, 2022). The two variations of SVM, via the use of Lagrange multipliers will ensure resilience to the phenomenon of overfitting, and also lead to sufficient generalization, especially on unbalanced data sets.

Random Forest Classifiers showed similar results to SVM, but their results were poorer than SVM, implying that they were not good at capturing complex nonlinear relationships as had been observed. This difference can be explained by the fact that Random Forest uses the concept of random subspace sampling, which is unlikely to take the full advantage of global statistically descriptive characteristics embedded in sparse TF-IDF vectors. Also, since the ensemble tended to overfit high-dimensional input without enlisting enough contextual knowledge, their recall had been slightly better, but their overall predictive performance was slightly less good.

The fact that this study has not factored in the utilization of contextual embedding models, like BERT, which are trained to describe word dependency and semantic context using transformer architectures, is also noteworthy. Although BERT has demonstrated the capacity to perform better than classical models in many NLP tasks, its absence in this study was not by chance since it would have made the research very computationally expensive and even uninterpretable. Nevertheless, the lack of context information might restrain the identification of subtle language patterns inherent in sarcasm, irony, etc., which might be a task better suited to transformer-based embeddings in the future.

6.3 Significance of Findings

The results of this study confirm the relevance of the practical and scientific significance of applying a combination of hybrid NLP and machine learning models to the classification of sarcastic headlines, which, in anticipation, may find extension to detect fake news. In a practical sense, using a Support Vector Machine (SVM) model and applying the Term Frequency Inverse Document Frequency (TF-IDF) feature extraction gives a computationally light and scalable option. It is especially favorable on platforms with low processing capability, like fact-checking widgets in a browser or a social media moderation console, where processing speed is essential.

On top of the growth of misleading information on the digital channels, this model architecture can be used to incorporate automated filtering of the potentially misleading information or sarcastic messages, thus helping to maintain the integrity of information (Alabid and Taher., 2024). TF-IDF vectors are also simple and interpretable, which promotes transparency of the model and enables explainable AI in practice, which is a critical requirement where applications need to be accountable, e.g. legal verification or a journalist verification system.

Scientifically, the findings substantiate the hypothesis that classical machine learning models do not lose competitiveness in performance once put in liaison with a robust feature engineering methodology. Although the model developed does not possess semantic awareness that exists in the TF-IDF, lexical and syntactic patterns were successfully exploited when used to differentiate between sarcastic and non-sarcastic headlines. This strengthens the efficacy of vector space models $V=\{v_1, v_2, \dots, v_n\} \mathbb{R}^d$ in which each document representation v_i is a frequency-weighted representation of tokenized text, and as such, are better suited to the linearly separable classification problems as done by SVM.

Nonetheless, the drawbacks of this method are distinctive. TF-IDF dependency disregards the sequential relationships among tokens as well as the contextual dependencies that are important in interpreting the subtleties of language structures like irony, sarcasm and misrepresentation. Therefore, there is also scope that future work can include distributional semantics by using word embedding methods like Word2Vec or GloVe, or through the use of context-based embedding, like via transformer-based models, such as BERT, or RoBERTa. Also, ensemble techniques can be more effective in generalizability and data performance across multiple linguistic datasets, such as model stacking or voting classifiers, which incorporate the outputs of both deep learning and classical ML approaches.

Chapter 7: Conclusion

The proposed research reached the successful challenge of identifying fake news and did so utilizing a combination approach that includes the use of Natural Language Processing (NLP) and classical Machine Learning (ML) classifiers. The original research question: How can hybrid NLP and ML methods be used to improve the accuracy and robustness of fake news detection programs? has been answered on the solid basis of the experimental work due to the preparation of the data, feature extraction based on vectors and testing of the models, allowing for a conclusion about their high accuracy and stability.

Combining techniques such as TF-IDF-based vectorization and the Logistic Regression, Support Vector Machines (SVM), and the Random Forest algorithms in a hybrid pipeline showed significant improvements in the performance metrics compared to their separate practice. In particular, the SVM demonstrated the best Area Under the ROC Curve (AUC = 0.92) and, thus, the best discriminating capabilities of the items of real and fake news (Raghavendra and Niranjanamurthy., 2024). This strengthens the strength of integrating both linguistic feature engineering and sound classification strategies in cases of binary text classifications.

The results of the study indicate that the feature extraction process using NLP, including lemmatized tokenize, stop-word filter and frequency-based vectorization, can be a significant addition in making the sparse text model more predictable and smoothing out the noise in text data. TC: The terms in which mathematical terms of the TF-IDF weighting are expressed:

$$TF-IDF(t,d)=TF(t,d).\log\left(\frac{N}{DF(t)}\right)$$

It has shown a good performance in converting text into a format capable of being represented in a way that makes sense to machines.

Unfortunately, the project has used a secondary dataset, consisting of labeled sarcastic and non-sarcastic headlines, which has proven capable of being used to reproduce misinformation modeling, whereas its syntactic and semantic similarity to fake news headlines is enough to make it of use in this case. The NLP-ML hybrid structure presented in this paper is not only large-scale and combinatory, but can also be used in real-time fake news detection pipelines.

The study therefore has a significant contribution in the field of understanding of natural language and automated mitigation of misinformation.

Chapter 8. Project Reflection

The project has been an all-out learning experience on how to implement the Natural Language Processing (NLP) and Machine Learning (ML) in a real-life scenario, fake news identification by parsing sarcasm and misleading headlines. The secondary dataset, which consisted of annotated headlines of high-quality sources, allowed the researcher to pay the entire attention to the modeling pipeline, optimization of the algorithm, and strategy of the system without coming to grips with a time-consuming process of gathering primary data.

Technically, it was necessary to implement the preprocessing of text, tokenization, lemmatization, and stop-word removal in order to normalize raw text data into a format that can be read by a machine. The development of several classifiers, such as Logistic Regression, Support Vector Machines (SVM), and Random Forest, aided in developing an insight into the behavior of an algorithm on each of the text-based features. The performance was compared in detail using the precision, recall, and F1-score and ROC-AUC as metrics that allowed a better focus on the best practice in the presence of a non-uniform class distribution.

The project faced a number of difficulties. Preprocessing and dealing with textual data were also inconsistent, especially about dealing with punctuation and edge cases in the process of lemmatization. In addition, the model hyperparameters needed to be tuned and cross-validated iteratively in order to obtain optimal results without overfitting them. The Random Forest model was computationally demanding in combination with high-dimensional TF-IDF vectors, but robust at the same time. Conversely, when operating on the vectorized dataset, SVM proved to generalize and be fast.

This project elucidated the shortcomings of a static approach to feature extraction and opened the possibility of strengthening that approach in future versions through the use of contextual embeddings, in this case, BERT. The inability of the existing system to be multilingual also limits the system to being defined globally; thus, the possibility of future work is to include multilingual transformer models and real-time APIs to implement.

On the whole, the project improved the expertise of the researcher in the end-to-end text classification systems, including data preparation, selection and construction of features, training models and visualization of their performance. It strengthened theoretical knowledge and practical experience that are very necessary in the data science application in NLP.

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Appendix

```
import pandas as pd
import numpy as np
import re
import string
import nltk
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split, cross_val_score,
StratifiedKFold
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.metrics import classification_report, confusion_matrix,
roc_auc_score, roc_curve, accuracy_score, precision_score,
recall_score, f1_score
```

```
nltk.download('punkt', force=True)
nltk.download('stopwords', force=True)
nltk.download('wordnet', force=True)
nltk.download('omw-1.4', force=True)
```

```
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
```

```
df1 = pd.read_json("/content/Sarcasm_Headlines_Dataset.json",
lines=True)
df2 = pd.read_json("/content/Sarcasm_Headlines_Dataset_v2.json",
lines=True)
```

```
df1['article_link'] = df1.get('article_link', None)
df1['headline'] = df1['headline'].astype(str)
df2['headline'] = df2['headline'].astype(str)
```

```
df = pd.concat([df1, df2], ignore_index=True)
df.drop_duplicates(subset='headline', inplace=True)
df.dropna(subset=['headline'], inplace=True)
```

```
stop_words = set(stopwords.words('english'))
lemmatizer = WordNetLemmatizer()
```

```
def preprocess(text):
    text = text.lower()
    text = re.sub(r"http\S+", "", text)
    text = text.translate(str.maketrans('', '', string.punctuation))
    tokens = word_tokenize(text)
    tokens = [lemmatizer.lemmatize(w) for w in tokens if w not in
stop_words and w.isalpha()]
    return " ".join(tokens)
```

```
df['clean'] = df['headline'].apply(preprocess)
```

```
vectorizer = TfidfVectorizer(max_features=5000)
X = vectorizer.fit_transform(df['clean'])
y = df['is_sarcastic']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, stratify=y, random_state=42)
```

```
models = {
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "Random Forest": RandomForestClassifier(n_estimators=100),
    "SVM": SVC(kernel='linear', probability=True)
}
```

```
results = {}
conf_matrices = {}
roc_curves = {}
```

```
for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    y_prob = model.predict_proba(X_test)[:, 1]
```

```
    results[name] = {
        "Accuracy": accuracy_score(y_test, y_pred),
        "Precision": precision_score(y_test, y_pred),
        "Recall": recall_score(y_test, y_pred),
        "F1 Score": f1_score(y_test, y_pred),
        "AUC": roc_auc_score(y_test, y_prob)
    }
```

```
    conf_matrices[name] = confusion_matrix(y_test, y_pred)
    roc_curves[name] = roc_curve(y_test, y_prob)
```

```
metrics_df = pd.DataFrame(results).T
print("\nModel Performance Metrics:\n", metrics_df)
```

```
metrics_df.plot(kind='bar', figsize=(12, 6))
plt.title("Model Comparison")
plt.ylabel("Score")
plt.xticks(rotation=0)
plt.ylim(0, 1)
plt.grid(axis='y')
plt.legend(loc='lower right')
plt.tight_layout()
plt.show()
```

```
for name, matrix in conf_matrices.items():
    plt.figure(figsize=(4,3))
    sns.heatmap(matrix, annot=True, fmt='d', cmap='Blues')
    plt.title(f"Confusion Matrix - {name}")
    plt.xlabel("Predicted")
    plt.ylabel("Actual")
    plt.tight_layout()
    plt.show()
```

```
plt.figure(figsize=(8,6))
for name, (fpr, tpr, _) in roc_curves.items():
    auc = results[name]['AUC']
    plt.plot(fpr, tpr, label=f"{name} (AUC = {auc:.2f})")
plt.plot([0,1], [0,1], 'k--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curves")
plt.legend()
plt.tight_layout()
plt.show()
```